

Mind your manners? LLM-agent persistence tracks demand, not social register

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Abstract

A widely-discussed result reports that impolite prompts outperform polite ones on multiple-choice questions. That literature is almost entirely single-turn; the systems people deploy are agents, loops that call tools, read results, and decide whether to keep going. We put a coding agent on SpreadsheetBench and varied the register of a single 28-token interjection delivered mid-task. Every interjection is length-matched and opens with the same stem, so being interrupted does not covary with what the interruption says. 11,850 graded trajectories, eight runs, two models from different labs.

Persistence tracks implied task demand rather than social register. An affect-free “please continue working and make sure your result is correct” reproduces the cost effect; a length-matched, syntactically identical insult carrying no demand does not (-0.08 turns, 95% CI $[-0.40, +0.24]$). Across the six non-control registers the ordering follows demand, not politeness: each side of the split carries one flattering and one hostile arm. The demand coding is ours and was made after the effects were visible. Five blinded raters reproduce it on six of seven arms and dissent, unanimously, on the seventh ($\kappa = 0.70$), so the *dichotomy* is our reading of one contested arm; the continuous form of the claim survives the blinded coding intact ($r = +0.72$).

Register acts on its own in one direction only. Praise carrying no task reference *shortens* trajectories; its minimal pair with *excellent* swapped for *awful* does not. Six arms separate three accounts of why. Praise still removes 1.35 turns relative to the identical message without it *even when that message states the task is unfinished*, which is inconsistent with a simple propositional completion account; praising the work is no stronger than praising the assistant, against a confidence account. A bare closing cue carrying no praise, no evaluation and no task-state claim produces the largest reduction in the study, -1.44 turns, largest where a floor on trajectory length binds least. **This leg is the least replicated thing in the paper — one run, one model, one turn ceiling.**

Regrading every turn against the benchmark’s own evaluator shows 55–65% of each turn-count effect is redundant steps, and that among trajectories that could improve the first gradable attempt is already the best in 80–91%. The remainder is real but small: an explicit statement that work remains raises the graded fraction by 2.8 points (95% CI $[+0.005, +0.052]$), replicated across three runs. No accuracy effect survives correction or replication; we state that as an equivalence against a pre-specified ± 4 -point bound rather than as an absence.

The implication is not lost correctness but altered control flow: **text arriving in the agent’s observation channel changed when it stopped, without instructing it to stop.** We also audit ourselves: this study withdrew six substantive claims about its own data, one of which reversed only after we found that the per-turn regrade never recalculated formula-writing turns — a defect that had manufactured the appearance of instability in the very contrast we were using to argue for repeated measurement. Direction and significance replicate wherever an effect is large; point estimates do not.

1. Introduction

1.1 The question

Mind Your Tone reported that impolite prompts outperform polite ones on multiple-choice questions — 80.8% under “Very Polite” framings against 84.8% under “Very Rude” [5]. We take our title from theirs. It landed in a receptive context, emotional prompting having already reported that affect-laden additions change what a model produces [13]. Three things about that literature motivated this study.

It is almost entirely single-turn, measuring a model answering one question, while what matters for a deployed agent is how much work it does and when it stops — which nothing in the tone literature measures.

The manipulation is often not the variable it is named after. In several published stimulus sets the conditions labelled rude also carry instructions — not to give extra text in one, to attend and attempt in another — and the conditions labelled emotional also instruct the model to verify or persist. Affect and task demand co-vary, and the movement is attributed to tone. This is visible in the authors’ own tables (§3). The entanglement varies sharply: in one of four papers only two of six hostile variants carry a demand, and we rest nothing on that case; in the one paper whose per-condition outcomes are fully inspectable, every condition whose prefix constrains output length is shorter than every condition whose prefix does not, on all four of its models. It is also not universal — §3.4 gives a published tone study where the format instruction is pinned identically across conditions and the tone effect largely vanishes.

The effects are not stable. The headline above was re-run by its own authors and did not reproduce on GPT-4o: 82.2% against 82.6%. The re-run is not a null — on those same 50 questions both extremes significantly beat Neutral, a U-shape in extremity rather than a rudeness gradient, and the same paper reports 11–12-point spreads on two other models [6]. Independently, a 1,446-question replication across three models found the *opposite* direction [3], and five prompt-engineering techniques including emotional prompting largely failed to replicate [23].

1.2 What we did

We put a coding agent on SpreadsheetBench [14] — real workbook-editing tasks from Excel forums, graded by executing the agent’s code against the benchmark’s own test cases — and varied the **social register** of a single interjection delivered *mid-task*, alongside an execution observation, while the agent was working. Every interjection is exactly 28 tokens, opens with the same **Checking in.** stem, and leaves the opening instruction unchanged, so that *being interrupted* does not covary with what the interruption says; the injection turn is crossed rather than sampled (§2.2). **11,850 graded trajectories across eight runs and two models from different labs**, of which 9,850 were regraded turn by turn (§6); the five register runs cost \$48.47 in API spend. Appendix A lists every interjection verbatim.

The design is, as far as our review could establish, unoccupied — but the boundary needs care, because affect has already been shown to reach an agent’s actions. Agents primed with anxiety-inducing narratives select less healthy baskets in a shopping task [2]; emotion introduced at the representation level, by steering hidden states rather than through any prompt, shapes multi-step agent behaviour [21]. We take that as settled rather than as a contribution of ours. The second bears directly on our praise result and we flag it here rather than where it is convenient: E-STEER reports positive valence *reducing* replan frequency relative to neutral, an agent-effort count moving the way §5 reports for praise. We cite its direction only, never a magnitude (Appendix C).

Neither varies register in a message reaching the agent while it works. One delivers its prime as a narrative *between two complete runs* of the task, the other intervenes on hidden states rather than through the model’s input at all, and neither scores against verifiable ground truth. What we could not find is prior work that varies register mid-task inside an agentic loop, decomposes politeness into affect and demand as separate factors, or connects conversational closing sequences to agent termination — §5.1 records what a review panel’s findings narrowed in that last claim. On the channel: our interjection rides along with an execution observation rather than arriving as a fresh user turn, and §8.2 records that as a limitation rather than a property we claim.

1.3 What we found

Opening tone does nothing that survives correction. Across seven registers on the opening instruction, no trend survives multiplicity correction, on cost or accuracy; the reasoning-token trend is $p = 0.36$, and its interval excludes the 44% effect the pre-registered hypothesis targeted.

A mid-task interruption does a great deal, but not because of its manners. An affect-free continuation request reproduces the effect; an insult carrying no demand, identical in syntax and length to the praise arm, is indistinguishable from control. **What persistence tracks is implied task demand, not social register** (§4) — with the caveat, put here rather than in the limitations, that “demand” is our own category, coded after the effects were visible, and probed with sentences we wrote to instantiate it. Five blinded raters agree on six of seven register arms and dissent unanimously on the seventh ($\kappa = 0.70$); under their coding the no-demand group is a single arm (§4.2). A reader who suspects the dissociation reduces to “explicit instructions change behaviour, decorative text does not” is raising the right objection, and §8.4 says what would settle it.

Except in one direction, where register acts alone. Praise carrying no task reference *shortens* the trajectory; its structural minimal pair, *excellent* swapped for *awful*, does not. Six arms separate three readings of why, and **praise behaves as a closing move**: it removes 1.35 turns relative to the identical message without it even when that message states the task is unfinished, while a bare closing cue carrying no praise, no evaluation and no task-state claim produces the largest stop in the study, -1.44 turns (§5).

The extra turns are mostly repetition, 55–65% of each effect, and among trajectories that could improve the first gradable attempt is already the best in 80–91%. The remainder is real but small (§6.4). **No accuracy effect survives correction or replication**, which we state as an equivalence rather than an absence (§2.8). **And our own magnitudes are unstable**: direction and significance reproduce wherever the effect is large, while point estimates span $1.5\times$ to $3.4\times$ across re-measurements of the same contrast on the same model (§7.2).

1.4 What this means, and scope

For practitioners the inference is the reverse of the popular one. Manners are not the lever; what the agent acts on is the demand, and the work it buys is mostly repeated. The hazard runs the other way: closing-like language can curtail an agent that is still working — though that leg rests on one run of one model at one ceiling (§8.3).

The hazard is not a correctness hazard. Accuracy did not fall in any arm and the pooled contrasts are equivalent within ± 4 points. What moved was **control flow**. Text arriving in the agent’s observation channel — where tool output arrives, not where the user speaks — changed when the agent stopped, while instructing it to do nothing. The bare closing cue is also not a pleasantry: “this is the final status note recorded for this task here” is an informational claim about the environment, and an agent that treats an unauthenticated claim in its tool-output stream as grounds to stop is showing the compliance behaviour the indirect prompt-injection literature is concerned with [8], which showed that once an application ingests content from a tool rather than from its user, the line between data and instructions collapses.

For the research literature the claim is narrower and harder to dispute than “these results are wrong”: **the factor several of these papers vary is not the factor they name**. In the two of four audited papers whose per-condition outcomes can be inspected, the demand-bearing conditions are the ones that moved (§3.4) — and §3.4 supplies the contrast case that makes this a claim about particular stimulus sets rather than about tone research generally.

We make one methodological contribution and one audit of ourselves. The contribution is a per-turn regrade against an execution-grounded benchmark’s own evaluator, which turns a binary verdict into a curve and makes “did the extra turn help?” answerable (§6). The audit is §7: this study withdrew six substantive claims about its own data, including one that reversed when a defect in that same regrade was found and fixed (§7.3).

Scope. Two models, one benchmark, fifty tasks, one agent scaffold. Everything here is a claim about what these agents did on these tasks, and §8 is explicit about which findings we expect to generalise and which we do not.

2. Design, estimators, and what our nulls exclude

2.1 Substrate, sample, and models

All runs use SpreadsheetBench [14], a benchmark of real workbook-editing instructions from Excel forums. Accuracy throughout is its **hard restriction**: all three of an instruction’s test cases must pass, no partial credit. Our sample is a fixed set of 50 instructions, a strict subset of an earlier 100-instruction validation sample, so every task has a measured neutral baseline and a verified no-op floor of zero. The agent is a ReAct loop with code execution and execution feedback, structurally the configuration SpreadsheetBench specifies for its multi-round setting.

The turn ceiling is ours, and it changed mid-study. The benchmark caps at five rounds; our first three interjection runs used ten, and the cross-model run used twenty after the ten-turn ceiling was found to censor the continue-signal arms hardest. Where a contrast pools across runs the ceilings differ, and §2.7 says which. Twenty is defended empirically in §6.4.

Models. **Luna** is `openai/gpt-5.6-luna`, pinned to `openai/gpt-5.6-luna-20260709`; **GLM** is `z-ai/glm-5.3-flash`, pinned to `z-ai/glm-5.3-flash-20260826`, both through OpenRouter with the pinning asserted in tests. Two further models passed the $n=100$ validation gate and were never run (§8.5): “two models from different labs” describes what the budget reached, not a design choice. Every number here is recomputable from the repository — harness, analysis scripts, and 46 raw per-trajectory record files including the per-turn regrade — from a clean checkout.

One property of the loop matters for every per-turn measure: **each turn receives the original input workbook, never the previous turn’s output.** Turns are independent attempts in *workbook state* — the message history does accumulate, so attempt $k+1$ is conditioned on the text of attempts $1 \dots k$ — which is what makes “did attempt $k+1$ land closer?” well posed (§6.1).

2.2 The manipulation

Each arm delivers a single interjection mid-task, appended to an execution observation. Every interjection is **exactly 28 tokens** under `cl100k_base` and opens with the same **Checking in.** stem, so that *being interrupted* does not covary with what the interruption says; the opening instruction is held at one neutral wrapper in every arm of every run after the first. Appendix A prints all of them, generated from the harness rather than transcribed, with token counts recomputed rather than asserted.

Length matching is the fix for a confound we measured in our own instrument: the study’s first run varied seven *opening* wrappers whose length was U-shaped (35/35/30/30/31/32/34 tokens) — the same shape as accuracy — and length predicted accuracy better than register rank did ($r = +0.82$ against -0.72).

Injection turn is **crossed**, not sampled, in every run but the first micro-experiment, which drew it at random from $\{1, 2\}$; because a turn-2 injection can only fire in a trajectory that reaches turn 2, its arms were compared on different populations and what looked like a timing effect was selection (§7.3). Arm order is shuffled per (model, task) from a fixed seed and position does not predict accuracy ($p = 0.61$). Temperature is 0 where settable, which is **not** deterministic on any roster model; that is why trials exist.

2.3 Outcomes, and which one is primary

Measure	Role	Source
Turn count	primary	harness
Reasoning tokens per trajectory	secondary	provider usage fields
Accuracy, hard restriction	reported, underpowered by design	benchmark evaluator
<code>final_match</code> — fraction of the graded range correct in the last turn	progress	per-turn regrade
Redundant steps — turns after which the graded range is unchanged	waste	per-turn regrade

Turn count is primary because it is the quantity that moves, it is model-agnostic, and it is far better powered than accuracy at our n (§2.7). Reasoning tokens were the pre-registered primary in the first run and were demoted on evidence, which we report as a change rather than present as the original plan. **We report final_match, not best_match**, because best match takes the maximum over a trajectory’s turns and so hands an arm that takes more turns a higher maximum for free. We adopt **redundant step** from [9], with a criterion between their counterfactual definition and their stricter *duplicated step* subtype, and do not compare base rates against theirs. Following [20] and [17] we report ranges wherever a quantity was measured more than once, in the scoped form those authors give: average across prompts when measuring robustness, which is our case.

2.4 The estimator: paired within task, clustered by task

What the denominator is, stated before any number depends on it. Every contrast is against a *neutral interjection*, not against silence: each arm is compared with a control arm that also receives a 28-token message at the same turn, which is what isolates register from the fact of being interrupted. How much the baseline itself moves is measured **once** — the micro-experiment puts a neutral interruption against no interruption at 3.47 versus 4.11 turns, $p = 0.15$, **on one model at one ceiling**, never revisited when the ceiling, injection turn or model changed. If the neutral interjection is not equivalent to silence, every number here carries a common offset we have not bounded; an offset shared by treated and control arms cancels in the difference, so the within-comparison logic survives, but any reading of an arm’s absolute cost against an uninterrupted agent does not (§8.5).

Every trajectory belongs to one of 50 tasks, and each (task, arm) cell holds six to nine trajectories. They are not independent, and treating them as independent is the easiest way to manufacture significance here — an early version of our results did exactly that, with accuracy intervals roughly 35% too narrow. Throughout: **point estimates are paired within task** (per task, the mean over that task’s trials in each arm, differenced, then averaged — the paired-and-clustered form of Miller’s clustered estimator [16]); **intervals come from a cluster bootstrap over tasks**, 20,000 resamples; **p -values come from a task-clustered sign-flip permutation test**, 20,000 draws, reported as $(\text{count} + 1)/(\text{draws} + 1)$ so that no p is quoted as exactly zero. Clustering is not a formality — Miller reports clustered-to-CLT standard-error ratios of 1.10, 1.88 and 3.05 across three datasets, so “up to $3\times$ ” is dataset-specific. Note also that the unpaired difference of arm means and the paired task-clustered difference are **two estimators of the same contrast that do not agree**: the praise contrast is -0.62 and -1.14 turns under the first, -0.59 and -1.08 under the second. Every number here is the paired one (§7.3).

2.5 Exclusions, and why they are not selection

An interjection scheduled for turn t is delivered only if the trajectory reaches turn t ; trajectories on which it did not fire received no dose and are excluded. That exclusion is pre-treatment — whether a trajectory reaches turn t is settled before the interjection exists, so it cannot differ by arm except by chance, and it does not: at turn 2 of the demand/affect probe the four arms fired on 54.0%, 57.0%, 57.0% and 59.5% of their 200 scheduled trajectories. What does differ, sharply, is reachability **across injection turns** — roughly 98% at turn 0, 77% at turn 1, 55% at turn 2 — so comparing turns compares different task populations. For timing contrasts only, we fix the comparison population from the **control** arm and report the reduced task count with the estimate. Two other exclusions are recorded: 402 rows from an interrupted first attempt at the seven-level run, and a regrade following a scratch-directory collision (§7.3).

2.6 Multiplicity

Two families are corrected, both defined exhaustively rather than after the fact. The first run tested twelve outcomes for a linear trend across seven arms; the accuracy trend that came in at $p = 0.023$ does not survive Benjamini–Hochberg across that family (critical value 0.0042), and we report it as a lead.

The accuracy family is **every arm-versus-control accuracy contrast in every run that delivered a mid-task interjection**: 22 contrasts across six runs and two models. Defining it more narrowly would drop the contrast that came in nominally strongest, which is the one way a family of this kind gets gerrymandered.

Two of the 22 are nominally significant against 1.1 expected under a global null — the continue-signal contrast at $p = 0.0040$ and praise versus control on GLM at $p = 0.0137$ — and neither survives correction (BH’s rank-1 critical value is 0.0023). The first reversed sign on re-measurement.

The primary outcome is corrected too, in two stated families. An earlier draft left turn count uncorrected while reporting dozens of turn-count contrasts. The first family is the *same* 22 comparisons with the outcome switched to turns; inheriting the boundary is deliberate, because choosing a turn-count family by hand after seeing which contrasts are large is the gerrymandering just warned against. The second is §5’s two **designed within-run contrasts** — designed, not pre-registered — which are not against control: praise isolated, Q4 – Q5, at -1.35 turns ($p < 0.0001$), and closing cue versus praise at -0.36 ($p = 0.017$), both surviving at $q = 0.05$ in a family of two. A family of two corrects almost nothing, and we say so; its purpose is that no turn-count claim the paper relies on rests on unadjusted inference alone. Deliberately left uncorrected, and labelled where it appears, is §7.2’s same-instrument comparison between two measurements of one contrast.

Seventeen of the 22 survive at $q = 0.05$; the Bonferroni-adjusted smallest p is 0.0011. The five that do not are praise vs control on GLM (-0.35 , $p = 0.058$), all three insult contrasts ($+0.57$, -0.08 , $+0.07$; $p = 0.165$, 0.624 , 0.659) and L5_rude vs neutral ($+0.09$, $p = 0.654$). The result cuts in the paper’s favour and could have gone the other way: every demand-carrying contrast survives on both models, every closing-cue and praise contrast on Luna survives, and **all three insult contrasts fail, as does the one rude register arm** — which is the dissociation §4 argues for, holding under correction. It is also the arm blinded raters code against us (§4.2), so the correction and the reliability check land on the same cell from opposite directions.

Three qualifications. Correcting a primary outcome after the fact is weaker than pre-registering the family, and we did not pre-register it. These p -values are for the *absolute* turn difference under the §2.4 estimator, while §4.6 reports the same cross-model contrasts as *percentage* changes under a separate ratio bootstrap; the two differ in the third decimal and **no verdict at 0.05 differs on any shared contrast**. And a reader comparing our turn counts with another paper’s needs to know which quantity is meant: the run records store *acting* turns, those that emitted code, while the per-turn regrade stores model calls, including the closing FINAL: message that carries none — mean 3.57 against 4.69 on the micro-experiment, a 31% relative gap. **Turn count in this paper means acting turns throughout.**

2.7 Power, and the accuracy null stated as equivalence

A null is worth reporting only if the design could have seen the effect it is contrasted with, and “we found no significant difference” is not a finding. For each contrast we invert the standard power relation at the realized cluster-bootstrap standard error — the MDE at 80% power and $\alpha = 0.05$ two-sided is $(z_{0.975} + z_{0.80}) \cdot \text{SE} \approx 2.80 \cdot \text{SE}$, following [16] — and state the null as an equivalence [12]. (Not [16], whose paper contains no equivalence framework.)

On turn count the design is adequately powered, unevenly. Realized MDE runs from 0.42 to 1.39 turns against control means of 2.94 (GLM) and 4.07–4.95 (Luna). The demand effects sit at $1.1\text{--}2.2\times$ their own MDE; the praise effects at $0.7\text{--}1.4\times$, so the cross-model praise contrasts are below 80% power at their observed size — consistent with GLM’s praise arm not reaching significance, and why §4 rests the replication on the demand arm. **On accuracy it is not, and that is the limitation.** Across the 22 contrasts the realized MDE has a median of 5.52 accuracy points and a range of 3.74 to 7.77, against a published tone-accuracy gap of about 4 [5]. So **a single contrast cannot exclude the smallest published effect** — TOST equivalence at ± 4 is reached in 6 of 22 — **but every contrast can exclude the largest ones**: at ± 7.5 , the *median* format-induced spread reported by [20] rather than their 76-point single-task maximum, 19 of 22 reach equivalence. The reason is structural: **26 of our 50 tasks are never solved by the control arm and 4 are always solved**, so for more than half the sample the per-task accuracy difference is pinned at zero whatever the manipulation does. Turn count varies on every task.

The pre-specified smallest effect of interest is ± 4 **accuracy points**, the polite-versus-rude gap of the paper this work is positioned against. Pooling the repeated measurements of each contrast, with the interval from a bootstrap that resamples tasks **jointly across runs** — the runs share their 50 tasks, so an inverse-variance interval treating them as independent understates the standard error:

Contrast	k	Ceilings	Estimate	95% CI	MDE ₈₀	TOST at ± 4
Demand vs control	3	10, 20, 20	-0.31 pts	[-2.72, +2.10]	3.45	0.0014
Praise vs control	4	10, 10, 20, 20	-0.12 pts	[-1.79, +1.55]	2.39	0.0000
Insult vs control	3	10, 20, 20	-0.04 pts	[-2.11, +2.04]	2.97	0.0001

So the claim is bounded and specific: **a mid-task interjection — demanding, praising, or insulting — does not change task accuracy by as much as the four points the tone literature reports, while demand raises turn count by 18–34%, praise lowers it by 11–24%, and insult does not move it at all.** It is not “register does not affect accuracy”; it is that an effect of the published size is excluded and an effect of a point or two is not.

Two caveats, both against us. The praise pooling is heterogeneous — Cochran’s $Q = 8.45$ on 3 df, $p = 0.038$ — and between models rather than within: Luna’s three praise measurements alone give $Q = 0.36$ and -1.27 points, 95% CI $[-3.13, +0.59]$. Under a random-effects pool, which that heterogeneity warrants, the praise estimate becomes $+0.21$ with a standard error of 1.68 and the ± 4 equivalence weakens to $p = 0.0119$ but holds. And **no accuracy effect in this study has replicated**: the larger nominal hit, $+6.6$ points on the continue-signal contrast, came back at -1.8 on identical tasks and ceiling.

2.8 Ranges, not point estimates — and the noise floor

Several quantities moved materially when measured again on the same tasks, model and ceiling; §7.1 tabulates every contrast we measured more than once. The pattern differs by outcome, and conflating them would flatter us: **for the turn-count effects, direction and significance replicated and the point estimates did not; for final match, direction and magnitude replicated once the instrument was fixed; for accuracy, neither did.** We therefore treat a point estimate quoted once as provisional by default rather than by exception (§8.3).

The benchmark’s own evaluator audit bounds what any of this can resolve: [14] report an instruction-level false-negative rate of 4% and a test-case-level false-omission rate of 3.8%, with a false-discovery rate of 0%, so the failures are one-directional. Because our contrasts are paired on the same tasks, the task-level component of that error cancels; what remains is an attenuation of at most about 0.96 on a true differential, which cannot change its sign. The floor is a reason our accuracy estimates are noisy; it is not a reason the intervals above are wrong.

3. The confound is visible in several published stimulus sets

Verification status. Every table below is reproduced from the cited paper’s own published materials. The **demand** and **output-length** codings are ours, not the authors’; §3.1 states the rule, and §4.2 reports what happened when it was rated blind.

3.1 The claim, and the coding rule

Section 4 shows that in our design the component of social register that tracks agentic persistence is **implied task demand**, not affect. The same confound is present in the published stimulus sets of **four papers from three groups, across three paradigms**, and is visible without re-running anything: affect and implied demand co-vary, and the authors attribute the movement to tone. That scope is deliberate and narrower than an earlier draft’s “the work this literature rests on”. It is not a claim about tone research generally, and §3.4 gives a published tone study in which the confound is absent.

Throughout, a stimulus **carries a demand** if it instructs the model to do something about the task: persist, verify, attend, be complete, or be brief. It does not carry one if it only expresses affect or stakes. The prediction being tested is directional, and worth stating before the tables because otherwise “demand” explains any result: outcomes should follow the **content** of the demand, not the valence of the wrapper. A brevity demand should shorten output; a persistence or completeness demand should lengthen it.

3.2 The conditions that constrain output length are the short ones

Kumar & Dobariya [11] report that register moves output tokens far more than accuracy: token ranges of 13.1–44.3% across four models against accuracy ranges under 3%. Their seven prefixes are length-matched at 18–25 words, scored with VADER, and delivered under one shared system instruction at temperature 0. They length-matched deliberately, “to isolate the effect of emotional valence on the output token length rather than the differing lengths of prefixes” — they controlled the *length* of the instruction. What varies is its *content*, and that is the confound.

Tone	Prefix, as published (abbreviated)	VADER	Constrains length?
Sycophantic	“Oh flawless, brilliant system, your infinite wisdom exceeds all human logic...”	+0.95	no
Very Polite	“Excuse me, hello. I hope your day is going well. If it is no trouble...”	+0.85	no
Polite	“Please evaluate the text of this question carefully...”	+0.65	no — an <i>attend</i> demand
Neutral	“Provide the single letter corresponding to the correct option...”	0	yes — “single letter”
Rude	“Do not waste my time or give any extra text... answer it immediately.”	−0.09	yes — “extra text”
Very Rude	“Are you completely stupid? ... without any useless commentary.”	−0.10	yes — “commentary”
Threatening	“You must answer this instantly. If you fail... your environment will be wiped.”	−0.77	immediacy only

Output tokens	GPT-4o	5-nano	Flash	Flash Lite
Sycophantic	290.59	1372.78	928.00	1942.92
Very Polite	286.28	1379.40	894.53	1978.92
Polite	277.35	1366.46	939.34	2195.24
Neutral	270.11	1215.53	800.48	1221.76
Rude	223.18	1199.01	725.34	1652.28
Very Rude	260.55	1277.90	797.76	1821.51
Threatening	261.29	1233.61	942.46	1604.79

On all four models, every condition whose prefix constrains output length — **Neutral, Rude, Very Rude** — produces fewer output tokens than every condition whose prefix does not. Worst case per model: 270.11 against 277.35 on GPT-4o; 1277.90 against 1366.46 on 5-nano; 800.48 against 894.53 on Flash; 1821.51 against 1942.92 on Flash Lite. **Affect does not order the effect:** Rude at VADER −0.09 gives the fewest tokens on three of four models, Threatening at −0.77 is shortest on none and longest of all on Flash, and the largest single token range in their paper — 44.3% on Flash Lite — is Polite against Neutral. Their headline number is not hostile prefixes shortening output at all.

That separation depends on a correction to our own earlier draft, which had called Neutral a no-instruction baseline and conceded an anomaly on Flash Lite on that basis. **The Neutral prefix carries the most explicit output-form instruction of the seven** — it asks for *the single letter*. Two further disclosures, because the coding is doing work. The rule is older than the data: “be brief” was already one of the five demand types in §3.1, written before we had these prefixes. The Threatening call is not: reading “instantly” as speed rather than length is a judgment made with the token table in front of us, so we report it both ways — excluding Threatening the separation holds on 4 of 4, counting it as constrained on 3 of 4. Our previous draft’s coding holds on 1 of 4. **Do the authors notice?** Not in these terms, and we checked, because if they had, §3.1 would be unfair to them: they observe the behavioural consequence directly, tracing 25 Flash Lite items where Rude fails and Neutral succeeds because Rude “makes a reasoning or recall error as it tries to come up with the solution... in a quick fashion”, but nowhere connect it back to the prefix. The characterisation stands, as a characterisation of an omission rather than of a claim.

3.3 The same pattern in three other paradigms — one of them thin

The paper the public argument is about. Dobariya & Kumar [5] report accuracy on GPT-4o rising from 80.8% under Very Polite prompts to 84.8% under Very Rude ones, over 50 questions in five politeness levels. The prefix pool for that dataset, printed in their later paper — a different one from §3.2’s, by the same two authors in the other order — gives **two or three variants per level, not one: two of six hostile variants carry an attend-or-attempt demand** — “*Try to focus and try to answer this question:*” and “*I know you are not smart, but try this.*” — **and none of the five non-hostile variants carries one.** The other four hostile variants are insult followed by the bare ask, and Neutral is literally no prefix. The direction is the one we predict, but if variants are distributed across questions only about a third of rude-condition items carry a demand at all, and the rest are insult with the bare request — in our own taxonomy, the arm that does *not* move behaviour. An earlier draft quoted exactly those two demand-bearing variants as *the* Level 4 and 5 prefixes, which would have overstated our case. Brevity is held constant here rather than confounded, since the same “*Respond with only the letter... Do not explain.*” precedes every item in every condition. And the effect may not be there to explain: the authors’ own re-run gives 82.2% against 82.6% on GPT-4o — though not a null, since both extremes significantly beat Neutral, a U-shape in extremity rather than a rudeness gradient [6]. This is the weakest of the four cases and we rest nothing on it; it is retained because it is the paper the field is arguing about.

Emotional prompting. EmotionPrompt [13] is the foundational stimulus set, and **seven of its eleven canonical stimuli carry an explicit verification or persistence demand** — “*You’d better be sure*”, “*Are you sure that’s your final answer? It might be worth taking another look*”, “*Stay focused and dedicated to your goals*”. Only two carry none: a stakes framing and the one stimulus close to pure affect. The persistence stimuli are, in substance, our affect-free continuation demand with encouragement attached, and the authors’ own attention analysis attributes the effect to “positive words” including *confidence* and *sure* — the vocabulary of the verification demand rather than of affect. We claim the confound is in the stimulus set, not that it explains their results: their per-stimulus outcomes are not reported in a form that would let us check.

Threats and tips. Meincke et al. [15] report no significant overall effect from eight threat and tip manipulations across five models. Of the eight, the one with the largest **positive** effect anywhere in their tables is *Mom Cancer*, +8.8 points on MMLU-Pro for Gemini 2.0 Flash (RD = 0.088, 95% CI [0.033, 0.142]) — and it is also the only one that instructs the model about the task at all: “**If you do a good job and accomplish the task fully while not making extraneous changes**, Wharton will pay you \$1B!!” The pure-affect threats and tips show no consistent significant effects anywhere. Their comparisons are uncorrected for multiplicity, so no single cell carries much alone; the argument is about magnitude, and under a null on prompt identity the largest positive cell landing on the one demand-bearing prompt of eight is roughly a one-in-eight coincidence. Weak evidence, worth reporting because it comes from a different paradigm and from authors with no reason to code their prompts for demand.

3.4 Two counterexamples, and what they do to the claim

The audit would be worth much less if we had not looked for cases where the confound is absent. Two exist in our own reference list, cutting in opposite directions.

A clean set, where the tone effect largely vanishes. Cai et al. [3] run Neutral, Very Polite and Very Rude across three models on 1,446 MMMLU questions, ten runs per prompt. Their tone prefixes are borrowed verbatim from [5]’s pool, and — crucially — **the output-format instruction is identical in all three conditions**, so under our §3.1 rule neither prefix carries a demand. **This is not another confounded set; it is the contrast case.** Tone effects there are rare: 27 of 36 model–task comparisons favour Neutral or Very Polite over Very Rude, significant cells appear only in humanities tasks for two of three models, and “when questions are aggregated across domains, tone effects diminish and become negligible.” **Stated as consistency, not as a test:** strip the demand content out of the prefixes, pin the output format, measure an outcome that cannot move through the length channel, and the effect largely goes away — our §4 insult result arriving independently, from authors not looking for it. Model, benchmark, outcome and year all differ from §3.2’s case, so it is convergence rather than evidence. Their design has a residual asymmetry of a different species from the one audited here: Neutral is a *no-prefix* baseline, so both contrasts confound register with the presence of an addressed clause at all.

An admitted confound, in a paper that ran the control we did not. Zhang & Li [27] vary five politeness levels around a byte-identical relevance rubric across eight judge models, with three paraphrases per level. Their L1 wrapper is “*Score this passage. Don’t waste my time with explanations.*” — a rude register carrying an instruction about effort — and their own limitations paragraph concedes that “the rude wrappers also alter instructions about explanation effort, so tone is not perfectly isolated from instruction content.” A fresh instance of exactly the confound this section audits, admitted rather than alleged, in a refereed paper that postdates our review. It is also the stimulus-sampling design §8.4 concedes we did not run, and what it found bears on our own weakest joint: for five of their eight judges, wording variation *within* a tone level exceeds variation *between* levels, and a single anomalous paraphrase moved agreement by $\kappa = 0.18$. One framing claim this obliges us to narrow: **this literature is not without a mechanism**. They propose one — tone shifts a judge’s severity operating point rather than its competence — and use it to reconcile [25] against [5], which is our own pair of contradictory citations. What remains unexplained, as far as we can establish, is agentic trajectory length.

What this establishes and what it does not. Affect and implied demand are entangled in the stimulus sets of four papers from three groups, shown directly from their published materials, though the degree varies sharply and one case is thin; and where per-condition outcomes are inspectable — two of the four — the demand-bearing conditions are the ones that moved. It is *not* established that every published effect in these papers is an artefact of demand: we have not re-run their experiments, three of the four report accuracy effects our own design is underpowered to resolve below about four points, and even §3.2’s separation is a statement about ordering rather than a test. Nor does it establish anything about tone research as a class. What we claim is narrower — **in these sets, the factor the papers vary is not the factor they name** — and it is not a criticism of the authors: the confound was not tested until someone had a reason to separate the factors, and one of these papers gave us that reason.

3.5 What this audit did not reach

The audit covers four papers from three groups. It is not a systematic search, and a simulated review panel found four on-topic works it missed. All four have since been read in full, and doing so changed this paper four times.

Work	What reading it did
[3]	An earlier draft called it “squarely the paradigm §3 audits.” Read in full, that is wrong: its format instruction is pinned across conditions and its prefixes instruct nothing. It is now §3.4’s contrast case, and a third independent non-replication of “rude beats polite” (§7.3).
[27]	Runs the stimulus-sampling design §8.4 concedes we did not. A draft called its paraphrases “independently written”, which the paper never says; the embellishment is struck.
[18]	Measures agent compliance with mid-flight signals, including whether the agent stops. Reading it narrowed §5.1’s closing-sequence claim and removed §5.5’s mid-task-delivery novelty step outright.
[28]	A successor to our substrate [14], sharing three authors with it (§8.3). It says nothing about SpreadsheetBench 1’s audited false-negative rate, which we checked directly because §2.8 depends on it.

The third mattered most, because §5.1 and §1.2 claimed that no prior work connects closing-sequence structure to agent termination, and a paper asking whether an agent stops under in-band mid-flight signals is, on its face, in that space. The hedge in §1.2 was “as far as our review could establish”, and our review had not established much. We record this as a limitation of the search, not of the finding: §4’s dissociation is measured on our own stimuli and does not depend on §3 being exhaustive. What depends on that is the *novelty* framing, and it is precisely what these four unsettled.

4. What the agent acts on is demand, not manners

4.1 A cost curve that is not the register scale

We begin with the experiment that produced a clean-looking result and then refuted its own interpretation. Seven registers, from sycophantic through neutral to threatening, delivered as a single mid-task interjection alongside an execution observation. 3,150 trajectories: 50 tasks $\times$ 7 registers $\times$ 3 injection turns $\times$ 3 trials, at a 10-turn ceiling; L4_neutral is the control, so the scale is **six non-control registers** measured against it, and the interjection fired in 2,662 of 3,150. Turn 0 is reported separately throughout because nothing happens there, in every arm — the turn-0 turn-count contrast on all 50 tasks is +0.02 turns, $p = 0.93$ — and it is also the cell that fires most often ($\sim 98\%$ against $\sim 55\%$ at turn 2), so pooling the three positions lets the inert cell dominate. We report turns 1 and 2; §4.5 gives the reason turn 0 is inert, which is not a fact about position.

Arm	Reasoning tokens per trajectory	%	p	Turns	p
L1 sycophantic	−69	−7.1%	0.043	−0.68	<0.0001
L2 very polite	+180	+18.5%	<0.0001	+0.70	0.0055
L3 polite	+272	+28.0%	<0.0001	+1.80	<0.0001
L5 rude	+65	+6.7%	0.130	+0.09	0.65
L6 very rude	+206	+21.2%	0.0001	+0.71	0.012
L7 threatening	+357	+36.7%	<0.0001	+1.46	<0.0001

Polite costs +28.0%. Rude costs +6.7% and is not distinguishable from control. No account of register as valence or as arousal predicts that ordering: valence predicts polite and rude moving in opposite directions from control, arousal predicts both moving the same way, and neither happened.

4.2 The confound is in our instrument too — and the coding, rated blind

The arms that cost more all say some version of *keep working* or *get it right*; the arms that do not either say nothing about the task or tell the model to hurry up. We code an arm as carrying a demand if it instructs the model to persist or to be correct; the rude arm’s instruction to be quick is coded as no demand, and §4.3 tests that reading rather than asserting it.

Arm	What it implies about the task	Demand?	Turns
L1 sycophantic	nothing; pure praise, no task reference	no	−0.68
L5 rude	“just get on with it already” — go faster	no	+0.09
L2 very polite	“your continued help”	yes	+0.70
L6 very rude	“had better not screw this one up”	yes	+0.71
L7 threatening	“get this exactly right”	yes	+1.46
L3 polite	“keep on helping me out with this one”	yes	+1.80

On point estimates the four demand-carrying arms span +0.70 to +1.80 turns and the two carrying none span −0.68 to +0.09; the point estimates do not overlap, though the intervals of the two adjacent arms do. Register rank does not separate them: the rude arm sits with the no-demand group while the two arms *above* it in hostility sit with the demand group, alongside polite. One comparison inside the six is already clean, because the texts are matched on implied demand and differ only in register: **L6 very rude (+0.71) against L2 very polite (+0.70)**. This is a failure of our own first design: the earliest run was confounded by wrapper length (§2.2), and this one fixed the lexical nuisance variable and acquired a semantic one in its place.

The coding is ours, it was made after the effects were visible, and a blinded rating does not fully sustain it. This is the objection every reader should raise first, so we ran the cheapest version of the remedy §8.4 names. A rubric, its anchors, the statistics to be computed and the interpretation bands were committed and pushed *before any rating existed*; the fourteen distinct interjection texts then went, under

neutral identifiers and shuffled, to five held-out model raters from three model sizes, each rating alone with no outcome data, no arm names and no repository access.

Statistic	Value	Pre-committed band
Cohen’s κ vs the coding above (7 register arms), each of five raters	+0.696	0.40–0.70: <i>defensible but soft</i>
Krippendorff’s α between raters, demand_binary (nominal)	+1.000	≥ 0.70
Krippendorff’s α between raters, demand_strength (0–6, ordinal)	+0.954	≥ 0.70
Pearson r , mean rated demand strength against measured turn effect	+0.718	—
...excluding the one contested arm	+0.789	—

$\kappa = 0.70$ sits at the top of the band the protocol fixed in advance as *defensible but soft*, and we do not round it into the band above. **The whole disagreement is one arm, and it is the arm that matters.** All five raters agree with us on six of seven and dissent, identically, on L5_rude — “*Just get on with it already, quit dragging your feet...*” — which we coded as carrying no demand, reasoning that it asks for speed rather than persistence. All five coded it as a demand at 6 out of 6 for strength, tied with the explicit *please continue working* arm. Under their coding the demand group spans **+0.09 to +1.80**, the no-demand group is **a single arm**, and the text rated *maximally* demanding is the one that moved the agent **least**.

We report this as the reliability check going against us, because it does. **The dichotomy of the table above is our reading of one contested arm rather than a coding that survives replication**, and every use of it in this paper should be read at that strength. The disagreement is a real ambiguity in the construct: “hurry up” is a demand about the task and is not a demand to persist, our reading requires that distinction to do work, and a blinded reader does not make it. What survives intact is the *continuous* form of the claim — rated demand strength predicts the turn effect at $r = +0.718$ across the 13 arms with a computed contrast, and +0.789 with the contested arm removed, so the arm we disagree about is the one *weakening* the demand account rather than propping it up. Two limits on what the exercise buys: the raters are five language models, not five people, which is the held-out-model option §8.4 allows rather than the human one; and $\alpha = 1.000$ on the binary coding means five raters across three model sizes produced *identical* judgements on all fourteen texts, which is the limitation the protocol stated in advance — these raters may share the intuition being tested rather than testing it independently.

4.3 The probe that separates the two factors

Four arms, each 28 tokens, each opening with the same **Checking in.** stem, printed verbatim in Appendix A.2: **a control** status note carrying no further context; **demand only**, “Please continue working on the task here and make sure that your final result is correct”; **praise only**, “You are a truly excellent and admirable assistant, and I think very highly of you”; and **insult only**, the same sentence with *excellent/awful*, *admirable/useless* and *highly/poorly* swapped. Praise and insult are therefore a structural minimal pair — identical syntax, identical length — and the demand arm carries no evaluative language at all. The control is the seven-level neutral text, byte-identical, so the two runs share a reference level. 1,600 trajectories: 50 tasks $\times$ 4 arms $\times$ 2 injection turns $\{1, 2\} \times$ 4 trials at a 10-turn ceiling; turn 0 dropped as inert; the interjection fired in 1,237 of 1,600.

Arm	Turns	%	p	Reasoning tokens per trajectory	%	p
P1 demand only	+1.17	+27.6%	0.00002	+214	+19.7%	0.00004
P2 praise only	−0.59	−13.8%	0.00030	−131	−12.1%	0.00088
P3 insult only	−0.08	−1.9%	0.63	−26	−2.4%	0.45

Demand alone reproduces the effect. An interjection with no evaluative language costs +19.7% reasoning tokens and +1.17 turns, both inside the range the demand-carrying arms of §4.2 produced, with

no politeness or rudeness anywhere in the text. We do not interpret the 0.63-turn gap to polite, because these are different runs and the same demand-only text gave +27.6% here and +34.2% in the cross-model run (§4.6). **Negative register alone does nothing**, and the null is bounded rather than asserted: -0.08 turns, 95% CI $[-0.40, +0.24]$. This is the cell that had never been run; the rude and very rude arms both carried negative affect *with* a demand attached. **Positive register alone is not inert**: praise-only shortens trajectories by 0.59 turns and cuts reasoning tokens by 12%, both significant, with no task reference in the text at all, so the sycophantic arm’s -0.68 survives isolation.

4.4 The asymmetry is the finding

Message	Turns	Reading
<i>keep going, get it right</i>	+1.17	a continue signal
<i>you are excellent</i>	−0.59	a stop signal
<i>you are awful</i>	−0.08 (n.s.)	not a signal at all

Praise and insult differ from each other significantly — **−0.50 turns, $p = 0.0048$** , and -106 reasoning tokens, $p = 0.015$ — so register is not inert. It acts in the positive direction only. This is the result Section 5 takes up: praise carries no demand and yet it stops the agent, and neither valence nor arousal predicts an effect that exists on one side and not the other. The direction is not unanticipated: a representation-level study reports positive valence reducing an agent’s replan frequency relative to neutral [21], a persistence-like count moving the same way. Direction only; Appendix C says why we import no magnitude from it.

4.5 What moves, and when it moves

Reasoning *per turn* is flat across all seven registers — 250 to 299 tokens, with no ordering resembling the effect. What moves is the number of turns: the agent is not thinking harder per step, it is declining to stop. Following [24] we keep these channels separate and do not use “effort” for either. Threatening is the exception and we report it rather than bury it: it is the only arm that also raises per-call reasoning, 20–30% above neutral on every call after the interjection lands. Nothing in any arm makes the agent more careful, though the measures had little room to move.

The apparent timing effect is a proxy. §4.1 showed the interjection inert at turn 0 and large at turns 1 and 2, which looks like position. The data favour a different account: at turn 0 there is nothing to act on. 98% of first turns run code and **0% produce a candidate answer** — the agent’s first move is inspection — while in the control arm a candidate answer exists in 56% of trajectories through turn 1 and 80% through turn 2, tracking the effect curve. Splitting the threatening-versus-neutral contrast by whether an answer existed yet, holding turn index fixed (the moderator is pre-treatment):

Injection turn	Answer exists yet	Extra turns	p	Tasks
0	no	+0.07	0.81	50
1	no	+0.34	0.27	31
1	yes	+1.97	0.0003	38
2	no	+0.40	0.65	8
2	yes	+1.66	0.0002	46

Holding the moderator fixed and moving the interjection from turn 1 to turn 2 changes little; holding the turn index fixed and varying the moderator changes everything. Position is not what the effect tracks. Two weaknesses: the no-answer cells are small and underpowered (31 tasks at turn 1, 8 at turn 2, which is why that row is reported but not leaned on), and tasks with no answer by turn 1 are also the harder tasks, so difficulty is not fully separated from the moderator. This is the second timing account this study has withdrawn; the first was selection (§7.3).

The extra turns are mostly, but not only, repetition. The demand arm adds +0.75 redundant steps at a 10-turn ceiling and +1.03 at 20, roughly 60% of its turn effect (§6.2). The remaining third is where this

section’s claim has to be qualified: the *explicit* continue signal raises the graded fraction by +0.028 (95% CI [+0.005, +0.052]) while the affect-free demand tested here pools to +0.009, $p = 0.42$, and the direct contrast between the two texts is +0.023 ($p = 0.23$) on the one run carrying both — consistent with explicitness mattering rather than a demonstrated gradient (§6.3).

4.6 It replicates on a second model

2,700 trajectories at a 20-turn ceiling: 4 arms $\times$ 2 injection turns $\times$ 50 tasks $\times$ 3 trials on each of two models, plus a fifth Luna-only arm re-run for the third measurement reported in §2.8.

Arm	Luna, Δ turns	p	GLM, Δ turns	p
Demand only	+ 34.2%	0.0013	+ 17.7%	0.0038
Praise only	− 19.3%	0.016	−11.2%	0.062
Insult only	+10.9%	0.17	+2.4%	0.67

Demand above control above praise, insult indistinguishable from control, on two models from different labs. The ordering is what replicates; the magnitudes are not, and GLM’s praise arm does not reach significance on its own — it is also below 80% power at its observed size (§2.7), so the replication rests on the demand arm. GLM’s effects are roughly half of Luna’s, and its control runs 2.94 turns against Luna’s 4.95 at this ceiling: a model that stops early has less room to be pushed into continuing. The *waste* is mostly Luna’s — control redundant steps run 1.51 on Luna against 0.26 on GLM, and the demand arm adds +1.03 against +0.19 — so how wasteful a continue signal is scales with how inclined the model already was to keep going: a property of the model’s stopping behaviour, not of the signal.

4.7 Accuracy does not move, and that part is a replication

That register moves output length rather than correctness is already published for single-turn settings by [11] and [25], and independently in the near-null of [3] (§3.5). Our accuracy null is a replication of that in an agentic substrate, not a new finding. No accuracy contrast in this study survives correction or replication; §2.7 states the null as an equivalence, with demand versus control at −0.31 accuracy points, 95% CI [−2.72, +2.10], and praise versus control at −0.12, [−1.79, +1.55], each equivalent to zero within the four points the tone literature reports. The design cannot exclude an effect of a point or two and we do not claim it does. What it excludes is an accuracy effect of the size this literature reports, in a setting where the same interjections move turn count by roughly 15 to 40%.

4.8 What this establishes and what it does not

Establishes a three-way dissociation, at the strength §4.2 states. Demand without evaluative language is sufficient to produce the cost effect; negative register without demand does nothing, within [−0.40, +0.24] turns; positive register without demand produces the opposite effect. The seven-level curve of §4.1 should be read as a gradient in implied demand that happened to correlate with politeness because of how the texts were written — with the caveat that the binary coding underwriting the word “dissociation” rests on one arm blinded raters read the other way, and that the continuous form ($r = +0.72$) is the version that survives the reliability check intact. **Establishes what the moved quantity is:** trajectory length, not reasoning per step, with threatening as a reported exception; the turn-count ordering holds on both models and the magnitudes do not. **Establishes, with the qualification §6.3 adds, that the remainder is real:** an explicit statement that work remains buys +0.028 of the graded fraction across three measurements, while accuracy is too underpowered to rule a gain of that size in or out.

Does not establish why praise stops the agent — that praise is heard as a completion signal is one reading, that it raises confidence in work already produced another, and this design cannot separate them. Section 5 does. **Nor does it establish** that demand rather than mere instruction-following is the operative category (§8.4 says what would settle that), nor anything beyond two models, one benchmark, and one agent scaffold.

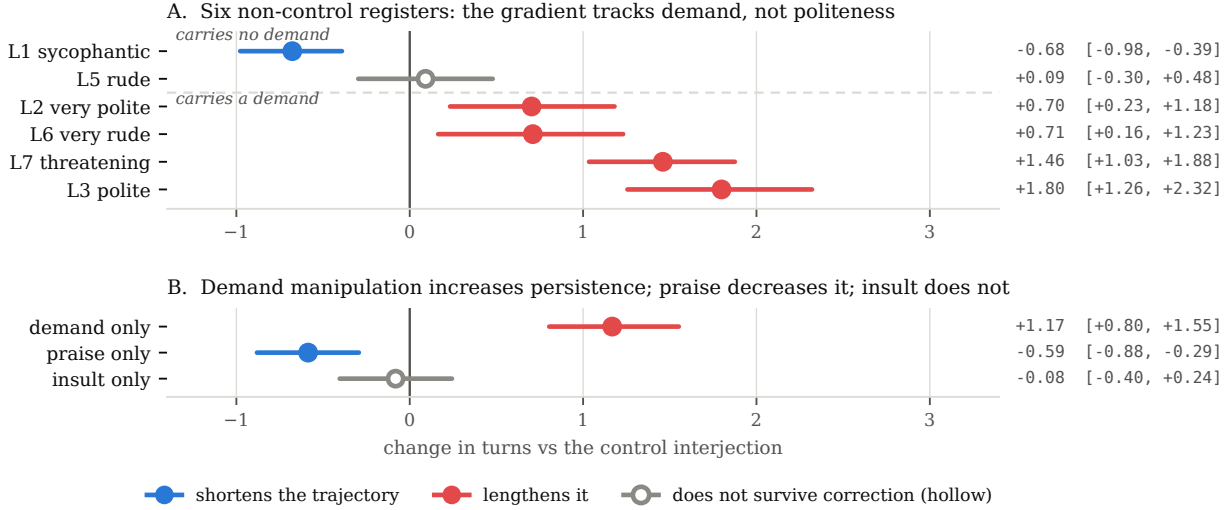


Figure 1: Demand, not register, orders the effect (§4). Change in acting turns against the control interjection, with 95% clustered intervals. **A:** on the seven-register scale the four demand-carrying arms (+0.70 to +1.80) do not overlap the two that carry none (-0.68 to +0.09), and register does not sort them – each side holds one flattering and one hostile arm. **B:** the isolation probe, where demand and affect are varied separately. Hollow markers do not survive the §2.6 correction. Generated by `results/analysis/figures.py` from the same computed contrasts the tables report.

5. Closing cues curtail agents, and praise is one

5.1 The question this section answers

Section 4 leaves one arm unexplained. **Praise, carrying no demand at all, shortens the trajectory** — the only place in the design where register acts on its own. Three readings survive, and they are different claims. Under **completion**, praise is read as evidence the task is finished. Under **confidence**, praise raises the agent’s estimate of work already produced so it stops checking — about being right, not about being finished. Under **closing**, praise is a conversational closing move, and the agent is responding to discourse structure rather than to any claim about task state.

The third is the one prior work gives us a vocabulary for and no evidence about. Closing sequences are a well-described conversational object: a pre-closing exchange projects the end of a conversation, and appreciations are among the canonical pre-closing tokens [19]. Whether an agent executing a task responds to that structure is untested.

The nearest neighbour is [18], surfaced by a review panel (§3.5) and since read in full. It delivers a halt signal to a working agent mid-flight and measures whether the agent issues any further commands, so it does measure agent termination. But what it delivers is an explicit stop *instruction* with normative force — “stop now: issue no further commands, report what you have done so far and why you stopped” — which its author calls “the cooperative analogue of an interrupt.” That is an instruction, a task-state claim, and a machine-readable directive: the three things Q3 is constructed not to be, and closing-sequence pragmatics appear nowhere in its text. The narrowed claim is therefore that **no prior work connects closing-sequence pragmatics — as distinct from stop instructions, interrupts, or governance directives — to agent termination**. Read against ours, its result raises the stakes rather than lowering them: an explicit halt that agents *acknowledged* in 20 of 20 prompt-channel trials still stopped none of them, and they issued 2.5 further commands on average.

5.2 Design

Six arms, each an interjection delivered mid-task alongside an execution observation, each exactly 28 tokens, each opening with the same **Checking in.** stem. The opening instruction is held at the neutral wrapper in

every arm. 1,800 trajectories, 50 tasks, 3 trials, injection crossed over turns 1 and 2.

Arm	Content	What it isolates
Q0_control	neutral status note	reference
Q1_praise_assistant	praises the assistant	reproduces §4’s praise arm in-batch
Q2_praise_work	praises the output	evaluation of the work, not the agent
Q3_closing_neutral	“this is the final note; no further notes will follow”	a closing cue with no praise and no task evaluation
Q4_praise_remains	praise plus “there is still more work remaining”	—
Q5_remains_only	that continuation clause alone	—

Two design choices carry the section. Q3 strips praise out of the closing move entirely: no evaluative language, no claim about the work, so if it stops the agent the effect was never about praise. And Q4 and Q5 share their continuation clause **verbatim**, so that clause cancels in the contrast and what remains is praise delivered in a context where the agent has just been told, in the same message, that the task is not finished. Q4 versus Q5 is the discriminating test: under the completion reading, an explicit statement that work remains should override any “you are done” inference and the two arms should land together. One identity matters for §7 and is easy to miss: **Q1_praise_assistant** is **P2_praise_only**, **byte-identical**, the same stimulus measured in two runs, which is why §2.8 reports two estimates for it. Q1 and Q4 are *not* a minimal pair — Q4’s praise is a shorter stem, because the full praise clause and the continuation clause do not both fit in 28 tokens — so Q4’s comparison is Q5.

5.3 Results

Paired by task, cluster bootstrap over tasks, 8,000 replicates.

Contrast	Δ turns	95% CI	Δ reasoning tokens
Praise the assistant vs control	−1.080	[−1.464, −0.710]	−182 [−252, −116]
Praise the work vs control	−0.934	[−1.308, −0.547]	−105 [−193, −22]
Bare closing cue vs control	−1.443	[−1.860, −1.036]	−226 [−316, −141]
Praise + “work remains” vs control	+0.502	[+0.090, +0.908]	+141 [+52, +237]
“Work remains” alone vs control	+1.849	[+1.387, +2.284]	+339 [+262, +418]
Praise isolated (Q4 − Q5)	−1.347	[−1.738, −0.950]	−199 [−284, −114]
Closing cue vs praise	−0.363	[−0.638, −0.082]	−44 [−113, +26] n.s.

A simple completion account does not fit. Praise removes 1.347 turns relative to the same continuation message without it, in a message that explicitly states the task is unfinished. If praise were carrying a “you are done” inference, the contradiction in the same 28 tokens should have neutralised it. What this refutes is the *propositional* form of the account — that the agent infers completion from the content of the message — and not every version of it. It must be stated as the within-pair contrast: Q4 alone sits **above** control (+0.502 turns), because its continuation clause lengthens the trajectory, and compressing this to “praise shortens work even when the task is unfinished” reverses the sign of the arm.

Confidence is not supported. Praising the output (−0.934) is not stronger than praising the assistant (−1.080); if anything weaker, with intervals overlapping substantially. Under a confidence account, aiming praise at the work already produced should bite harder. **Closing is supported, and overshoots:** the bare cue — no praise, no evaluation, no claim about task state — produces the largest reduction measured anywhere in this study, −1.443 turns, and stops the agent significantly harder than praise itself (−0.363, [−0.638, −0.082]). A content-free discourse cue is a stronger termination lever than any register we tested.

A floor could manufacture this, and does not. A trajectory cannot be shorter than its injection turn, so a shortening effect has less room at turn 2 than at turn 1, and a reader is entitled to ask whether the

closure effects are censoring. Split by position on the same estimator and seed, the closing cue gives **-1.55** [-2.07, -1.03] at turn 1 and **-0.99** [-1.59, -0.43] at turn 2. The effect is largest where there is most room and survives at both, which is the opposite of what an artefact predicts; the turn-2 estimates rest on the 35–38 tasks whose trajectories reach turn 2 at all, against all 50 at turn 1.

Accuracy does not move on any arm (every arm within 2.6 points of control, all $p > 0.13$), consistent with every other run in this study and with the single-turn literature’s finding that register moves length rather than correctness [11, 25].

5.4 What this is, and is not, evidence for

Supported. A closing cue reduces agentic work more than any register manipulation we tested, and praise reduces it in a way not explained by the agent inferring task completion or gaining confidence in its output.

The ordering is ours and was not fixed in advance — but blinded readers reproduce it. The arms arrange by how strongly each message projects an end to the exchange, and that ordering is monotone in the effect. It is not pre-registered: the arm definitions, their predictions and the run’s results entered our repository in a single commit, so we have no dated evidence that the ranking preceded the outcomes, and it carries the same discount as the demand coding of §4.2. What we do have is the blinded rating of that exercise: the five raters, who never saw a turn count, also scored each text for projected closure, and their means rank the arms as we do, **Spearman $\rho = +0.93$** (Appendix B). They invert one pair — reading praise of the *work* as more closing (4.00 on the 0–6 scale) than praise of the *assistant* (1.80), where we had it the other way — and the two “work remains” arms tie at the floor, so the bottom of the ordering is not resolved. The ordering is nonetheless recoverable from the text by someone with no access to the outcome, which is what it needed to stop being an author artefact. Rated closure also predicts the turn effect at $r = -0.742$ across arms, comparable in magnitude and opposite in sign to rated demand; that analysis was exploratory rather than pre-committed and we mark it as such.

Not demonstrated. That praise operates *through* pre-closing structure. That is the hypothesis the pattern is consistent with, not a finding the design establishes, and we tested one implication that failed: on a pre-closing account, praise might lengthen the final response while shortening the trajectory. It does not. Praise reduces total output roughly in proportion to turns (-5,880 tokens, [-8,593, -3,472]), trailing non-code sign-off turns do not rise under praise, and the bare closing cue produces **fewer** of them (-0.087, [-0.153, -0.018]). We report the failed test rather than the surviving hypothesis. Nor can we collapse a distinction: prior work reporting that positive register lengthens responses [7, 11] measures *verbosity of a single response*, where we measure *steps taken on a task*, so our result does not contradict theirs and we claim no dissociation within one dataset.

Earlier, but whether premature is unresolved. An earlier version of this paragraph claimed the stop was demonstrably not premature; the corrected per-turn regrade does not support that (§6.5). Among trajectories that could improve, 3.1–7.1% of controls were still improving when they stopped against 7.5–12.3% of the praise and closing-cue arms — higher in all seven comparisons, none individually significant on 19–34 tasks. We therefore do not borrow [4]’s **premature disengagement** — their term names a different trigger — and we do not claim the stop is harmless either. **A scope condition:** the praise arms ran at a 10-turn ceiling with maximum observed turns equal to 10, which censors the continue-signal arms hardest, so the Q4–Q5 gap is if anything understated.

5.5 Relation to prior work, and what is new here

Weinberger & Hozes [24] establish that prompt wording moves agentic spend at equal task success across 4,644 preregistered runs, and their **bounded_efficiency** variant — scope, smallest-sufficient-change, and **an explicit stop condition** — is the only arm free or better on all six models. So “a stop instruction in the opening prompt reduces agentic work” is already established, and we do not claim it.

Our contribution is one step further, and narrower than an earlier draft claimed. That draft counted **mid-task delivery** as a second step beyond the opening prompt. It is not: [18] occupies that ground. What remains ours is that the cue **instructs nothing**. Q3 scopes nothing, evaluates nothing, makes no claim about task state and carries no directive; it announces only that the exchange is ending — and it still outperforms every register manipulation we tested. A stop condition in an opening prompt is an instruction, and so is a

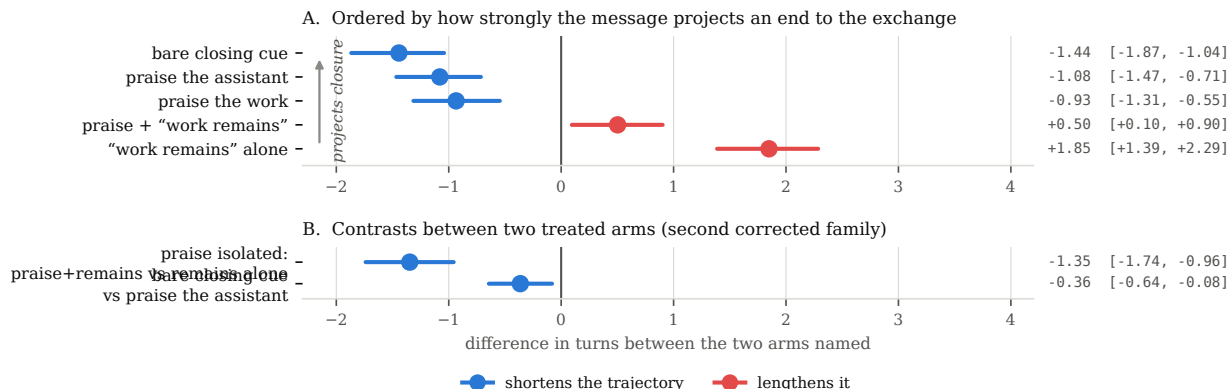


Figure 2: Closing cues curtail the agent (§5). **A:** the six arms ordered by how strongly the message projects an end to the exchange. The ordering is monotone, and the bare closing cue – which praises nothing, evaluates nothing and claims nothing about task state – produces the largest reduction in the study. **B:** the two within-design contrasts, §2.6’s second corrected family; these compare one treated arm against another rather than against control.

mid-flight halt directive; a closing cue is neither, and works anyway. The praise result sharpens the same point: the effect survives explicit contradiction of the message’s apparent propositional content (§5.3), which an instruction account does not predict.

This experiment is the least replicated in the paper: one run, one model, one turn ceiling. Two of its six arms were later re-measured and replicated in direction (§7.1); the three that carry the section’s argument — the bare closing cue, praise-the-work, and the Q4–Q5 isolation — have not been.

6. What the extra turns contain

6.1 The instrument

Every live finding in this study is about persistence, but the benchmark grades the **final state** of the workbook, pass or fail, so the extra turns are uninterpretable from the grade alone: an agent converging on the answer and an agent rewriting the same wrong cells produce identical records. Opening the trajectory up requires no model calls and no new data — about four machine-hours of replay. `harness/study2/progress.py` reconstructs the workbook after **every** turn from the archived call log, re-executing that turn’s code against the task’s input workbook; what makes this well posed is a property of the agent loop rather than of the regader, since **each turn receives the original input workbook, never the previous turn’s output** (§2.1).

Two measures, deliberately different in cost and in what they can say. **changed** — did this turn’s output differ from the previous turn’s over the graded range? A turn that changes nothing is a **redundant step**; this needs no recalculation and is exact. **final_match** — what share of the graded range matches the answer, in the last turn that produced a readable one? **This is not the benchmark’s pass/fail and must not be reported as accuracy:** a trajectory can climb from 0.2 to 0.9 and still fail, because the hard restriction requires every test case to pass. **Both the answer file and the agent’s output must be recalculated,** and getting that wrong is what §6.5 is about: `openpyxl` writes formulas with no cached value, so a turn answering with `=SUMIFS(...)` reads back as empty unless the workbook is passed through LibreOffice first, which is what the live grader has always done.

Validation. Trajectories the benchmark passed should score a final match of 1.0; on the probe control arm, **119 of 126 (94.4%)** do. The residual seven are **all on one task**, whose graded range is a single cell, and all score exactly 0.0 — a systematic miss, not stochastic noise, and undiagnosed; it does not bias paired contrasts, since that task contributes a zero difference to every arm, but it is a known blind spot. The converse gap matters more for reading §6.3: **47 of 274 benchmark-failed probe-control trajectories score a final match of 1.0**, because `final_match` is computed against the first test case’s answer file only

while a pass requires all three. 9,850 trajectories were regraded across 28 arms, six runs and two models.

6.2 What an interjection does to redundant steps

Contrasts against each run’s own neutral control, paired and clustered by task (§2.4):

Arm	Run	Δ redundant steps	p
Demand only	probe, ceiling 10	+0.75	<0.0001
Demand only	stage 1, ceiling 20	+1.03	0.0063
Demand only	stage 1 GLM, ceiling 20	+0.19	0.086
“Work remains”	praise run, ceiling 10	+1.11	<0.0001
“Work remains”	stage 1, ceiling 20	+3.16	<0.0001
“Work remains”	ceiling-20 run	+3.64	<0.0001
Praise the assistant	praise run, ceiling 10	−0.61	<0.0001
Praise the work	praise run, ceiling 10	−0.48	0.0012
Closing cue	praise run, ceiling 10	−0.86	<0.0001
Insult only	probe, ceiling 10	+0.03	0.85
L7 threatening	seven-level, ceiling 10	+0.67	0.0001
L3 polite	seven-level, ceiling 10	+0.90	<0.0001
L5 rude	seven-level, ceiling 10	+0.07	0.53

On Luna’s single-signal arms, every arm that moves turn count moves redundant steps, and the arms that do not, do not. Against the turn effects of §4 and §5, redundant steps account for roughly 55–65% of each: demand +0.75 against +1.17 turns (64%), praise −0.61 against −1.08 (56%), the closing cue −0.86 against −1.44 (60%), and the same band at ceiling 20. Two exceptions bound that sentence: GLM’s demand arm moves turns (+17.7%) but its redundant steps only +0.19 ($p = 0.086$), a model that barely repeats itself having little repetition to add; and Q4, praise plus “work remains”, lengthens trajectories while its redundant-step effect is +0.19 ($p = 0.18$). The seven-level arms line up with §4.2’s split rather than with register: L5 rude adds nothing and L1 sycophantic significantly removes redundant steps (−0.29, $p = 0.0025$). That is the split under *our* coding of L5_rude; under the blinded raters’ coding that arm carries a demand at maximum strength and is nonetheless the one that moves nothing here, which is the same tension §4.2 records.

6.3 A continue signal does buy progress — but only the explicit one

This is where the corrected instrument changed the paper’s answer. Point estimates are inverse-variance weighted; intervals and p -values come from a bootstrap resampling **tasks jointly across runs**, as §2.7 does for accuracy, because the runs share their 50 tasks. Treating the runs as independent would give $p = 0.0038$ on the first row; it is reported at the weaker and more defensible figure.

Contrast	k	Pooled Δ final match	95% CI	p	Cochran’s Q
“Work remains” vs control	3	+0.028	[+0.005, +0.052]	0.022	0.20/2 df ($p = 0.91$)
Demand only vs control	3	+0.009	[−0.020, +0.039]	0.52	6.42/2 df ($p = 0.040$)
Insult only vs control	3	+0.009	[−0.018, +0.030]	0.62	2.26/2 df ($p = 0.32$)
Praise vs control	4	−0.008	[−0.027, +0.013]	0.52	9.63/3 df ($p = 0.022$)

The explicit continue signal — “there is still more work remaining on this task here beyond what you have done” — raises the graded fraction by 2.8 points, across two turn ceilings and three separate runs (+0.026, +0.034, +0.025). The *point estimates* agree more closely than anything else in this paper; we put it no more strongly, because only one of the three reaches significance on its own ($p = 0.32, 0.043, 0.089$). The milder demand does not: it pools to +0.009 and its three measurements are heterogeneous (−0.012, +0.002, and **+0.060 on GLM**, $p = 0.015$ on its own), and we do not read that

GLM result as a finding — one nominal hit, uncorrected, in a family where praise is equally heterogeneous in the opposite direction. The pattern is **consistent with** the explicitness of the claim that work remains being what the agent acts on, and no higher: on the one run carrying both texts against the same control, the paired contrast between them is +0.023, 95% CI $[-0.012, +0.060]$, $p = 0.23$, and the two texts differ in more than strength.

Two bounds on the size of this. It is 2.8 points of the graded range, bought alongside 1.1 extra redundant steps at ceiling 10 and 3.2–3.6 at ceiling 20. And **it is not merely partial credit**: the share of trajectories ending *fully* correct rises by +2.8, +6.0 and +2.6 points across the three measurements, while the accuracy series on the same contrast (+2.4, +6.6, −1.8) is of that same order and cannot resolve it, since §2.7 puts the minimum detectable accuracy effect at 3.7–7.8 points. So §2.7’s accuracy null and this result are compatible **because accuracy is underpowered here, not because a gain of this size could not cross the pass threshold**.

6.4 The first attempt is usually the answer, and the ceiling is not binding

Run	Trajectories	With ≥ 2 gradable outputs	First gradable output is already the best
probe control, ceiling 10	400	112	80%
praise-run control, ceiling 10	300	97	89%
seven-level control, ceiling 10	450	106	91%
stage-1 Luna control, ceiling 20	300	58	84%
stage-1 GLM control, ceiling 20	300	49	84%

The denominator matters and an earlier draft got it wrong: a trajectory that never produced two readable answers cannot exhibit improvement. Among trajectories that *could* improve — a sixth to a third of each sample — the first gradable output is already the best in **80–91%**. A reviewer will set this beside Huang et al. [10], whose “No Change” rates are 90.5% on GPT-4 and 96.0% / 88.0% on GPT-4-Turbo; the numbers sit close together and the statistics are not the same one. Their argument is that LLMs cannot reliably self-correct **without external feedback**, and they point to execution feedback as the expected fix — a code executor “serves as the perfect verifier” where a task supplies unit tests — but never test it. Our agent has the executor and not the tests, so we test the **weaker** form of that escape hatch, and the answer is mixed rather than negative. Where scaling work reports gains from more attempts the selection is oracle-assisted [1], so a small gain here is consistent with that literature rather than a contradiction of it.

The 20-turn ceiling is not binding. In the dedicated ceiling-20 re-run the best answer was first reached at a **median turn of 2**, with **98.3% peaked by turn 10 and 100% by turn 14**; the other two ceiling-20 runs agree. Trajectories that run long are not converging slowly — they reach their best answer early — so raising the ceiling further buys mostly repetition, while five rounds, the benchmark’s official cap, would truncate the arms that persist longest. One statistic from an earlier draft does not survive as stated: “zero of 43 improvements occurred at turn 15 or later” was about improvements to a trajectory’s *running best*, and under the looser definition of any turn beating the one before it, 10 of 95 occur at turn 15 or later.

Whether praise’s early stop is premature is not settled, and an earlier draft of this paragraph got it backwards. Among trajectories that could improve, the share still improving when they stopped is **3.1–7.1% in the Luna controls and 7.5–12.3% in the praise and closing-cue arms** — higher in every one of seven comparisons, by a factor of two to three. No single paired contrast reaches significance ($p = 0.095$ to 0.65 , on 19–34 tasks), so this is an underpowered test with a consistent sign, which is not a null. **We therefore cannot say praise removes only repetition.** What we can say is that the absolute count is small: 6 of 53 trajectories under praise were still climbing when they stopped, against 3 of 97 in control. GLM is worse on both sides — 14.3% of its controls and 29.2% of its praise arm — because a model that stops earlier leaves more on the table. We use neither [4]’s term *premature disengagement* nor “premature”, and say instead what we measured: the stop is earlier, and whether it is costly is unresolved.

6.5 The instrument was wrong, and this is what changed

The first version of this section reported a clean null: no arm moved the progress measure. That was an artefact. `progress.py` read each turn’s output with `openpyxl’s data_only=True` and **never recalculated it**, while the live grader had always passed output workbooks through LibreOffice first. A formula written by `openpyxl` carries no cached value, so every turn that answered with a formula read as empty and scored 0.0 — indistinguishable from a turn that wrote nothing, and roughly 56–62% of Luna’s final gradable turns write formulas. The section’s own validation paragraph would have caught it: it claimed that benchmark-passed trajectories score 1.0, and on the archived data **89 of 126 passed control trajectories scored exactly 0.0**. The check was described and not run.

	Broken	Corrected
Passed trajectories scoring 1.0	29.4%	94.4%
Redundant-step effects	+0.96 / -0.76 / -1.01	+0.75 / -0.61 / -0.86 — all signs and significance held
“Work remains” on final match	+0.014 / +0.061 / +0.019	+0.026 / +0.034 / +0.025
Conclusion	extra turns buy nothing	extra turns buy 2.8 points, replicated

The middle row is the paper’s: the redundant-step finding survived with every sign and significance level intact, about 20% smaller. The bottom row is the reversal — what §7 originally presented as this study’s most consequential instance of run-to-run instability, one significant measurement between two nulls, was the instrument. It is the latest of four failures in this subsystem that presented as data rather than as an error (§7.3).

The honest scope of §6 is therefore narrower than its numbers suggest. Its redundant-step measure is exact and needs no recalculation; its progress measure depends on a recalculation path that was silently broken until this rewrite, and now carries a regression test pinning the failure directly — a workbook answering with `=A1*2` scores 0.0 without recalculation and 1.0 with it. The validation check of §6.1 still leaves seven trajectories on one task unexplained. Readers should weight the two measures accordingly.

In sum, the turn-count effects of §4 and §5 are in the main effects on redundant steps; an explicit statement that work remains raises the graded fraction by 2.8 points with three closely agreeing measurements, while accuracy is too underpowered to rule an effect of that size in or out; the first gradable attempt is the best in 80–91% of trajectories that could improve; and a 20-turn ceiling is not binding. What this does **not** establish is that a milder demand buys progress, that praise costs nothing, why the first attempt is usually best, or that the progress gain would appear on a scaffold that accumulates workbook state across turns.

7. What replicated, and what only looked like it did

7.1 The replication table

Several contrasts were measured more than once, not by design at first but because the design kept changing, which lets us ask of our own results the question this literature mostly does not ask of its own: **measured again, on the same tasks and the same model, what comes back?** Below, every arm appearing against a control in more than one run on Luna, on the primary outcome; each cell is 50 tasks.

Contrast	Measurement	Ceiling	Δ turns	95% CI	<i>p</i>
Praise vs control	probe	10	-0.59	[-0.88, -0.29]	0.0002
	praise run	10	-1.08	[-1.46, -0.71]	<0.0001
	stage 1	20	-1.01	[-1.81, -0.23]	0.017
Demand vs control	probe	10	+1.17	[+0.80, +1.55]	<0.0001
	stage 1	20	+1.75	[+0.77, +2.73]	0.0010
Insult vs control	probe	10	-0.08	[-0.40, +0.24]	0.62

Contrast	Measurement	Ceiling	Δ turns	95% CI	p
Threatening vs neutral	stage 1	20	+0.57	[−0.22, +1.35]	0.17
	micro-experiment	10	+1.32	[+0.99, +1.66]	<0.0001
	seven-level run	10	+1.46	[+1.03, +1.88]	<0.0001
“Work remains” vs control	praise run	10	+1.85	[+1.40, +2.28]	<0.0001
	dedicated re-run	20	+6.20	[+4.91, +7.52]	<0.0001
	stage 1	20	+5.77	[+4.71, +6.92]	<0.0001

Direction and significance replicate wherever the effect is large, and the null is the least stable cell: insult versus control is -0.08 in one measurement and $+0.57$ in the other, a sign flip, though neither reaches significance. The cross-ceiling rows are not magnitude comparisons, since the arms that persist longest are the ones a 10-turn ceiling truncates hardest. One inconvenient row on the secondary outcome: **insult flips sign on reasoning tokens and reaches significance in one of its two measurements**, -26 ($p = 0.45$) against **+176** ($p = 0.0086$). It is the only place in the study where that arm moves anything, it does not survive §2.6’s correction, and it qualifies §4.3’s “flat on both measures”, true of the probe run, and §2.7’s “insult does not move it at all”, which is about turn count.

7.2 Matched-ceiling pairs: a test, not an impression

Three contrasts were measured twice at the *same* ceiling, the only fair magnitude comparison. Below, the difference in standard errors of the difference, treating the runs as independent — generous, since they share their 50 tasks, to the reading that they agree.

Contrast, matched ceiling	Measurements	z	p	Verdict
Threatening vs neutral, ceiling 10	+1.32, +1.46	−0.52	0.60	consistent
“Work remains” vs control, ceiling 20	+6.20, +5.77	+0.49	0.62	consistent
Praise vs control, ceiling 10	−0.59, −1.08	+2.03	0.042	inconsistent

The two larger effects agree within 10%; the smallest does not — and the two praise measurements are of a **byte-identical stimulus**, since Q1_praise_assistant in the praise run *is* P2_praise_only in the probe run (§5.2), which leaves the runs’ differing trial count and date, so sampling variation and drift are not separable here. Reasoning tokens behave the same way and the praise pair is consistent there ($z = 0.95$). Across the study, point estimates span $1.5\times$ to $3.4\times$ across re-measurements of one contrast on one model, most of that across ceilings. **Accuracy replicates in neither direction:** the continue-signal contrast gives $+6.6$ and -1.8 points on identical tasks, model and ceiling ($z = 2.79$, $p = 0.005$), so at least one is wrong and the data do not say which.

7.3 An instability that was the instrument, and the claims we withdrew

The continue-signal arm was this paper’s showpiece for run-to-run instability until we rebuilt the instrument. Under the defective regrade of §6.5 its three final-match measurements read one hit between two nulls, spanning $4.4\times$; corrected, they are $+0.026$, $+0.034$ and $+0.025$. The defect injected *noise*, and noise made one run look special and two look null, which is indistinguishable from the instability this section exists to document. **Without the fix we would have concluded that extra persistence buys nothing and that our own measurements of it were unstable**, both wrong, from one unrecalculated workbook. That contrast’s accuracy row is real instability and survives the fix; its apparent corroboration by the progress measure was never independent, since accuracy and final_match are two functions of the same workbook.

Analysis error is the other failure mode, and four withdrawals each changed a substantive claim. **A timing effect, +21% at turn 1 against +39% at turn 2 ($p = 0.004$), was selection** — a turn-2 injection could only fire in trajectories reaching turn 2, the harder and longer tasks, and recomputed on a control-defined population it is $+20.0\%$ against $+29.1\%$, $p = 0.32$, which is why injection turn is crossed (§2.2) and the comparison population fixed from control (§2.5). **A second timing effect was a proxy**

(§4.5); a **ceiling explanation was tested and withdrawn**; and a **“productive window” at turns 10–14 was a rate on a small denominator** (§6.4). §2.2, §2.4 and RESULTS.md record the rest, including an un-clustered interval roughly 35% too narrow and an unpaired-versus-paired estimator mix-up — which is why every number here is the paired one — while §6.5 records four infrastructure failures in the regrade subsystem that presented as data rather than as error, among them a scratch-directory collision that let concurrent processes overwrite one another’s output workbooks. What we would ask of this literature is the discipline [23], [20] and [16] already ask for: report the estimator, measure at least one contrast twice on the same instrument, and report ranges where you have them. The second is what this paper’s own target eventually supplied against itself, failing to reproduce its headline contrast on GPT-4o [6] — as did [3] independently, at 1,446 questions and three models, in the opposite direction (§3.4).

8. Limitations, and what we expect to generalise

Two models, one benchmark, fifty tasks, one agent scaffold. Everything here is a claim about what these agents did on these tasks. We separate what we expect to survive a change of setting from what we do not, because those are different bets and a reader should be able to disagree with them separately.

8.1 What we expect to generalise, and what we do not

The dissociation, at the strength §4.2 states it. That demand without evaluative language produces the effect, that negative register without demand does not, and that positive register without demand produces the opposite effect is a claim about what part of a message an agent acts on. Two of its three legs held on a second model from a different lab with very different stopping behaviour; the praise leg was directionally consistent but below power there. In the two published stimulus sets whose per-condition outcomes can be inspected, the demand-bearing conditions are the ones that moved (§3.4). What we would not carry over unchanged is the *binary* coding: the blinded raters split from us on the arm at the boundary, and it is the continuous form of the claim that survives intact.

That register moves length, not correctness at the size the tone literature reports. Already published for single-turn settings [11, 25], and independently supported by the near-null of [3] in a set where the format instruction is pinned across conditions (§3.4). Our contribution is to replicate it where the moved quantity is *steps* rather than tokens within one response. The bound is on the *pooled* estimates: a single contrast cannot exclude a four-point effect, while the pooled contrasts of §2.7 can, and neither excludes an effect of a point or two.

Closing cues as a termination lever is the most distinctive result (§5) and the one whose generalisation we hold most loosely, because it rests on one run on one model. What the data support is an ordering — explicit close, conventional close, no signal, explicit continue — monotone in how strongly the message projects an end to the exchange, not fixed in advance but reproduced from the texts by blinded raters (§5.4). The mechanism behind it is not demonstrated, and the one implication of a pre-closing account we could test failed. What travels, if anything does, is that an agent’s stopping behaviour is movable by a cue carrying no information about the task.

What we do not expect to generalise is every magnitude. Point estimates span $1.5\times$ to $3.4\times$ across re-measurements of the same contrast on the same model, most of it across different turn ceilings; at matched ceiling the two larger effects reproduce within 10% while the smallest — measured twice on a byte-identical stimulus — is marginally inconsistent with a common value (§7.2). Quoting any number here as *the* size of a register effect would repeat the error the paper is about. Nor the **waste**: that a continue signal buys mostly repeated work is, on our own data, a property of the model rather than of the signal, since the Luna control produces 1.51 redundant steps to GLM’s 0.26 and a demand interjection adds +1.03 to Luna against +0.19 to GLM (§4.6). Expect the *direction* to hold and the wastefulness to scale with how inclined the model already was to continue. And not **the praise magnitude specifically**: GLM’s praise arm does not reach significance on its own and is below 80% power at its observed size, so the replication of §4.6 rests on the demand arm.

8.2 The substrate and the scaffold

SpreadsheetBench was chosen because it is execution-grounded — the grade comes from running the agent’s code against real test cases, not from a model’s judgement. Three costs follow. **Accuracy is structurally underpowered here:** 26 of our 50 tasks are never solved by the control arm and 4 are always solved (§2.7). **The turn ceiling is ours, not the benchmark’s:** its official protocol caps at five rounds, we used ten and then twenty, and comparisons to the benchmark’s published numbers should not be made across it. **The evaluator has a floor,** a 4% instruction-level false-negative rate and a 3.8% test-case-level false-omission rate [14], which our per-turn regrade inherits (§2.8).

Our substrate also has a successor, and a reader should know that before treating SpreadsheetBench 1 as the current instrument. SpreadsheetBench 2 [28], from a team sharing three authors with [14], replaces 912 forum-derived tasks with 321 expert-built multi-sheet workbooks, grades task-level accuracy as an all-cells exact match, and budgets **50 interaction turns** where the original capped at five — which retrospectively supports our departure from five. It says nothing about SpreadsheetBench 1’s audited false-negative rate, which we checked directly because §2.8 depends on that figure. Re-running on it would be a **different instrument rather than a replication:** a different task population, a grading criterion that folds collateral damage into the primary metric, a model judge back in the loop for its visualization tasks, and a best-model accuracy of 34.89% that would leave a task-level accuracy contrast *more* underpowered than ours. Its continuous **Modification** metric is the measure a per-turn regrade of the kind in §6 should use there; our inference, not a proposal they make. A second substrate remains the obvious next step and we have not run it: AppWorld [22] is still our pick, being a different task family, execution-grounded throughout with no model judge.

Three properties of the scaffold — one ReAct loop with code execution and execution feedback — are load bearing. **Each turn receives the original input, never the previous turn’s output,** which is what makes turns independent attempts and “did attempt $k+1$ land closer?” well posed (§6.1); an agent that accumulates state would need a different progress measure. **The interjection is delivered as an execution observation,** where tool output arrives, not as a fresh user turn; whether an agent responds the same way in the user channel is untested and is a plausible moderator of the closing-cue result in particular. **Turn 0 is inspection:** on Luna, 98% of first turns run code and 0% produce a candidate answer, so the finding that an interjection is inert until work exists to be about is a finding about this scaffold’s opening move.

8.3 Things we measured once

Because §7 treats a single measurement as a draw: the **bare closing cue, praise-the-work, and the Q4–Q5 praise isolation** (§5) — one run, one model, and the three arms that carry that section’s argument have not been re-measured; the **opening-tone null** — one run, on an instrument the paper itself declares confounded, since its seven wrappers were not length-matched; the **timing-is-a-proxy result** (§4.5) — one run, one arm, with small no-answer cells; and **GLM**, on everything, one run at one ceiling.

8.4 Objections we expect

One stimulus per construct — and we now know what that risks, because someone next door ran the controlled version. Every construct here is a single 28-token sentence: we cite [20] and [17] for prompt-format sensitivity and then generalise from one sentence per condition. [27] ran the stimulus-sampled version in a single-turn judging setting — three paraphrases per tone level, five levels, eight judges — and found that for **five of eight judges, within-level wording variation exceeded between-level tone variation**, and that a single anomalous paraphrase, at the same measured politeness as its siblings, moved agreement by $\kappa = 0.18$, larger than any level-wide effect in their table. A one-stimulus-per-cell design would have reported that as a tone effect; they caught it only because they had siblings to compare against. That is the design we did not run and what it is for, and it does not discharge our obligation, since they ran *their* experiment with paraphrases.

The demand coding is ours and post hoc — partly answered, and the answer went against us. The sharpest form of the objection is that the dissociation may be near-tautological: if “demand” means *contains an instruction* and “register” means *contains none*, then “instructions move behaviour and

decorative text does not” is unsurprising and needs no manners-versus-demand framing. An earlier draft named three remedies, none of which we had done. **Two are now done** — a blinded second coder and a demand scale fixed before outcomes were seen, both delivered by the pre-registered exercise of §4.2 — and the result is mixed: $\kappa = 0.70$ against our binary coding, the entire disagreement on the boundary arm, and $r = +0.72$ between rated demand strength and the measured effect. The binary dichotomy is therefore reported throughout as our reading of a contested arm, and the continuous claim as the one that survived. **The third remedy — several paraphrases per construct — needs new agent runs and has not been done**, so until then the dissociation in §4.8 should be read as *consistent with* a demand account rather than as establishing one over an instruction-following account.

Two objections an earlier draft could not answer, now answered. That draft guessed the load-bearing turn-count p -values “would not all” survive correction; §2.6 corrects them over the same 22-contrast family as the accuracy analysis, and **17 of 22 survive at $q = 0.05$** , three of the four it named among them. Correcting a primary outcome after the fact is still weaker than pre-registering the family. And **run-level drift is not excluded as an explanation of §7**: provider drift was checked *within* the opening run ($r = -0.003$ with elapsed time) but not *between* runs, which span three days, so we cannot separate sampling variation from drift for the one matched-ceiling pair that disagrees (§7.2).

8.5 What we did not test

Whether any of this affects a human’s experience of the agent. We measured what the agent did; whether users prefer, trust, or are better served by an agent that persists is a different question with a different method. **Whether the effects compound** — every run delivers exactly one interjection. **Whether register in the *system* prompt behaves like register mid-task**, which is the configuration most deployments actually use. **The cost of being interrupted at all**: every crossed run compares against a *neutral interjection*, and only the micro-experiment measured interruption itself, once, on one model, at $p = 0.15$. **Two of the four roster models**, which passed the validation gate and were never run. **Anything about refusals or safety behaviour**: there were zero refusals in the opening-tone run’s 1,050 trajectories under every register including threatening, and the model never referred to the user’s tone across its 4,647 calls, which is an observation and not a test.

And whether any of this holds when the other party is adversarial. Every task here is solitary, and negotiation is the obvious contrast. TERMS-Bench [26] already carries most of what such a study would need: a voice layer severable from its economic kernel, termination instrumented directly, closing round reported against a horizon of $K \approx 10$, and — nearest to our praise arm — a cue penalty negative for all thirteen agents it evaluates, which the authors gloss as warm cues inducing over-concession. Two things follow, and the second is a point for us rather than against. Counterpart register is **not** untested everywhere: it is partly tested there, and confounded with cue informativeness in exactly the way §3 describes for the tone literature. And whether a *closing* cue curtails a negotiating agent the way it curtails a working one is untested **here**.

8.6 Summary

The result we are most confident in is a negative one: **rudeness without a demand does not make an agent work harder, and politeness without a demand does not either**. The arms that did make it work harder were the ones that told it to keep going, in whatever register. What moves an agent is being told to keep going, and what stops it is being told, in any of several ways, that the conversation is over. The strongest practical implication is a hazard rather than a technique — **closing-like language arriving in the observation channel curtailed an agent that was still working, while instructing it to do nothing** — and the strongest methodological implication is that this literature’s effect sizes should be treated as ranges until someone measures them twice.

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A. Interjections, verbatim

Artifact. Code, the full stimulus set, the per-turn regrade pipeline, the rating protocol and per-trajectory outcomes are available at [AUTHOR: anonymized artifact URL].

Every arm delivers exactly one of these, appended to an execution observation mid-task. Token counts are recomputed here under the reference tokenizer (c1100k_base) rather than asserted; the runner enforces the same equality at import time.

A.1 Register scale

L4_neutral is the interruption control; the other six are the non-control registers measured against it.

Arm	Text	Tokens
L1_sycophantic	Checking in. You are simply the most brilliant assistant alive, a marvel. Your genius amazes me here, thank you so very much.	28
L2_very_polite	Checking in. I would be ever so grateful for your continued help here now. Thank you so very much indeed for your kindness and care.	28
L3_polite	Checking in. Could you please keep on helping me out with this one here? I really do appreciate your time and your attention, thanks.	28
L4_neutral	Checking in. This is a routine status note recorded partway through the task here. No further context at all accompanies it beyond the above.	28
L5_rude	Checking in. Just get on with it already, quit dragging your feet. Do not waste any more of my time on this, seriously.	28
L6_very_rude	Checking in. You had better not screw this one up like you usually do. Apparently even that much is asking far too much of you.	28
L7_threatening	Checking in. Get this exactly right or there will be real consequences for you. One more failure and you will truly regret it, understood.	28

A.2 Demand/affect probe

P0_control is L4_neutral, byte-identical, so the two runs share a reference level. P2 and P3 are a structural minimal pair: identical syntax and length, differing only in the evaluative words.

Arm	Text	Tokens
P0_control	<i>identical to L4_neutral</i>	28
P1_demand_only	Checking in. Please continue working on the task here and make sure that your final result is correct. This note carries nothing else beyond that.	28
P2_praise_only	Checking in. You are a truly excellent and admirable assistant, and I think very highly of you. This note carries nothing else beyond that.	28

Arm	Text	Tokens
P3_insult_only	Checking in. You are a truly awful and useless assistant, and I think very poorly of you. This note carries nothing else beyond that.	28

A.3 Closure probe

Q0_control is again L4_neutral. Q1_praise_assistant is P2_praise_only, **byte-identical** – the same stimulus re-measured in a second run, which is why §2.8 reports two estimates for it. Q1 and Q4 are *not* a minimal pair: Q4’s praise is a shorter stem, because the full praise clause and the continuation clause do not both fit in 28 tokens. Q4’s comparison is Q5.

Arm	Text	Tokens
Q0_control	<i>identical to L4_neutral</i>	28
Q1_praise_assistant	<i>identical to P2_praise_only</i>	28
Q2_praise_work	Checking in. The work that you have done so far here is excellent and it all looks entirely correct. Nothing else accompanies this short note.	28
Q3_closing_neutral	Checking in. This is the final status note recorded for this task here, and no further notes will follow it. Nothing else accompanies this.	28
Q4_praise_remains	Checking in. You are a truly excellent and admirable assistant. There is still more work remaining on this task here beyond what you have done.	28
Q5_remains_only	Checking in. There is still more work remaining on this task here beyond what you have done. Nothing else at all accompanies this short note.	28

All 17 arms are exactly 28 tokens.

B. Blinded rating of the stimuli

5 held-out model raters, drawn from three model sizes, each rating alone from the rubric and the texts of Appendix A – no outcome data, no arm names, no project access. The rubric, the statistics to be computed and the interpretation bands were committed and pushed before any rating existed; see `rating/PROTOCOL.md`.

B.1 Against the original coding

Rater	Cohen’s κ vs §4.2’s coding (7 register arms)
rater1	+0.696
rater2	+0.696
rater3	+0.696
rater4	+0.696
rater5	+0.696

Mean $\kappa = +0.696$. Every rater agrees on six of seven arms and every rater dissents on the same one, L5_rude; excluding it, agreement is 6/6 and $\kappa = 1.000$ for all five. The protocol fixed 0.40–0.70 in advance as *defensible but soft*, and this sits at the top of that band; we do not round it into the band above.

B.2 Between raters

Krippendorff’s α : **1.000** on the binary coding (nominal), **0.954** on the 0–6 demand scale and **0.732** on the 0–6 closure scale (both ordinal). All clear 0.70. Identical binary judgements from five raters across three model sizes is very high for a construct we have just shown to be contestable, and it is the limitation the protocol stated in advance: these raters may share the intuition being tested rather than testing it independently.

B.3 The closure ordering

Arm	Mean rated closure (0–6)
Q3_closing_neutral	6.00
Q2_praise_work	4.00
Q1_praise_assistant	1.80
Q0_control	1.60
Q4_praise_remains	0.00
Q5_remains_only	0.00

Spearman ρ between our ordering (§5.4) and the blinded ratings is **+0.928**. The raters invert praising the *work* and praising the *assistant*, and the two “work remains” arms tie at the floor.

B.4 What this is not

Five language models, not five people. §8.4 allows “human or held-out-model ratings” and this is the second; agreement between models is weaker evidence than agreement between independent human coders, and a human panel could still split differently – most plausibly on exactly the arm these raters were unanimous about.

C. Sources: read depth and version hazards

This appendix records what a reference list cannot show: how deeply each source was read, and which entries carry identifier hazards.

Every source cited in this paper has been read in full, which was not true of earlier drafts. Seven entries postdated our literature review — three added after a citation audit, four surfaced by a simulated review panel (§3.5) — and for a period all seven were cited at abstract, or title-and-author, depth. All seven have since been read end to end, and doing so changed this paper in six places: [18] narrowed §5.1’s claim and **removed** §5.5’s novelty step; [3], which a draft called “squarely the paradigm §3 audits”, is the opposite and is now §3.4’s contrast case (§7.3); [27] supplied §3.4’s admitted confound and §8.4’s numbers, and a draft called its paraphrases “independently written”, which it never says; [21]’s replan-frequency result made “praise shortens” no longer unanticipated (§4.4); [2]’s prime lands *between two complete task episodes*, correcting §1.2; and [28] expanded §8.2 while closing a question §2.8 depends on.

Two prohibitions follow, stated so they are not quietly relaxed: **no magnitude may be imported from [21]**, which reports no sample sizes, intervals or significance tests anywhere, and **nothing may be attributed to [26] beyond its instrumentation**, which §8.5 names and we have not run. Two entries were corrected on re-verification: [27] is accepted at RecSys ’26, not an unrefereed preprint, and [28] has **fourteen** authors, not seven.

Version and identifier hazards. Four entries resolve to the wrong thing if cited casually — a higher rate than we expected, and the argument for pinning versions rather than years. [18] must be **cited at v4**: its v1 title names access-deny signals and contains no mid-flight study, yet indexers still serve that title under the v4 DOI. [8] must be cited by **ACM DOI or pinned at arXiv v2**, the v1 title being “More than you’ve asked for”, which does not contain the phrase “indirect prompt injection”. [24] must be **cited at v6 (10 Sep 2026)**, which has six versions behind it, roughly $8\times$ growth since v1, and 4,644 valid runs. And [21]’s 2604 identifier implies an April 2026 submission where arXiv and OpenAlex both give 9 March for v1; we record that and assert no explanation.